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Important concepts, symbols, and equations

- **forward dynamics** (for simulation): $\theta, \dot{\theta}, \tau \rightarrow \ddot{\theta}$
- **inverse dynamics** (for control): $\theta, \dot{\theta}, \ddot{\theta} \rightarrow \tau$
- two equivalent approaches to computing the dynamics:
 - Lagrangian (variational, based on energy)
 - Newton-Euler (“ $f = ma$ ” for the rigid bodies)

Important concepts, symbols, and equations (cont.)

Lagrangian approach:

Lagrangian

$$\mathcal{L}(\theta, \dot{\theta}) = \mathcal{K}(\theta, \dot{\theta}) - \mathcal{P}(\theta)$$

kinetic minus potential energy

eqs of motion

$$\tau = \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\theta}} - \frac{\partial \mathcal{L}}{\partial \theta} \in \mathbb{R}^n$$

$$\tau_i = \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\theta}_i} - \frac{\partial \mathcal{L}}{\partial \theta_i}$$

in components

Important concepts, symbols, and equations (cont.)

Standard forms of dynamic equations:

$$\begin{aligned}\tau &= M(\theta)\ddot{\theta} + h(\theta, \dot{\theta}) \\ &= M(\theta)\ddot{\theta} + c(\theta, \dot{\theta}) + g(\theta) \\ &= M(\theta)\ddot{\theta} + \dot{\theta}^T \Gamma(\theta) \dot{\theta} + g(\theta) \\ &= M(\theta)\ddot{\theta} + C(\theta, \dot{\theta}) \dot{\theta} + g(\theta)\end{aligned}$$

$+ J^T(\theta) \mathcal{F}_{\text{tip}}$

$M(\theta)$

$g(\theta)$

$c(\theta, \dot{\theta})$

$\Gamma(\theta)$

$C(\theta, \dot{\theta})$

$n \times n$ symmetric positive definite **mass matrix**

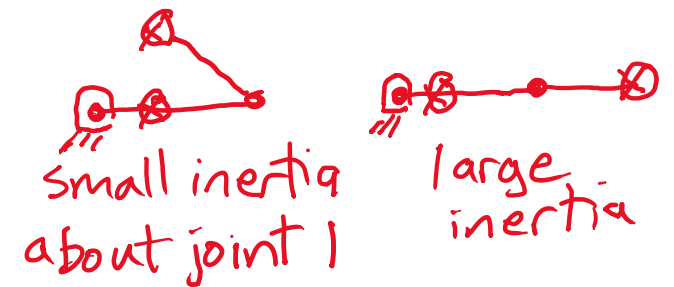
gravity (potential) terms

velocity-product terms

$n \times n \times n$ tensor of **Christoffel symbols** (due to nonzero $\partial M / \partial \theta$)

Coriolis matrix

depends on config θ !



Important concepts, symbols, and equations (cont.)

Velocity-product terms ($c(\theta, \dot{\theta})$, $C(\theta, \dot{\theta})\dot{\theta}$, $\dot{\theta}^T \Gamma(\theta)\dot{\theta}$) consist of

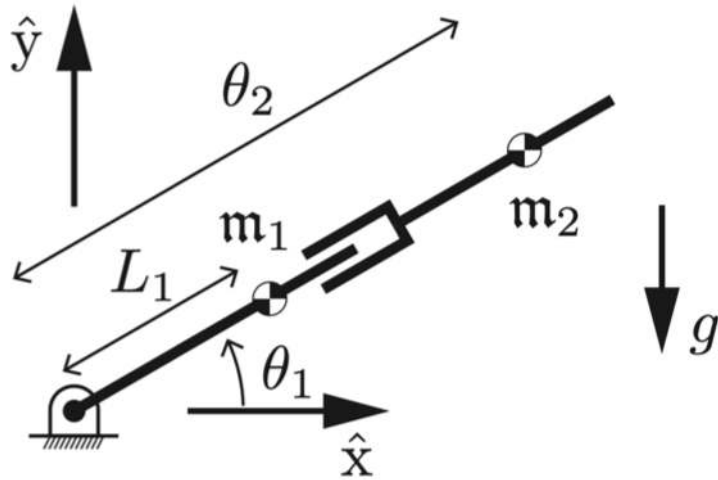
- **centripetal** terms, e.g.,

$$m_2 L_1 L_2 \dot{\theta}_1^2 \sin \theta_2$$

- **Coriolis** terms, e.g.,

$$-m_2 L_1 L_2 \sin \theta_2 (2\dot{\theta}_1 \dot{\theta}_2)$$

RP in gravity



$$\mathcal{P}_1 = \mathfrak{m}_1 g y_1 = \mathfrak{m}_1 g L_1 \sin \theta_1$$

$$\mathcal{P}_2 = \mathfrak{m}_2 g y_2 = \mathfrak{m}_2 g \theta_2 \sin \theta_1$$

$$\mathcal{K}_1 = \frac{1}{2} \mathfrak{m}_1 (\dot{x}_1^2 + \dot{y}_1^2) + \frac{1}{2} \mathcal{I}_1 \dot{\theta}_1^2 = \frac{1}{2} (\mathcal{I}_1 + \mathfrak{m}_1 L_1^2) \dot{\theta}_1^2$$

$$\mathcal{K}_2 = \frac{1}{2} \mathfrak{m}_2 (\dot{x}_2^2 + \dot{y}_2^2) + \frac{1}{2} \mathcal{I}_2 \dot{\theta}_1^2 = \frac{1}{2} \left((\mathcal{I}_2 + \mathfrak{m}_2 \theta_2^2) \dot{\theta}_1^2 + \mathfrak{m}_2 \dot{\theta}_2^2 \right)$$

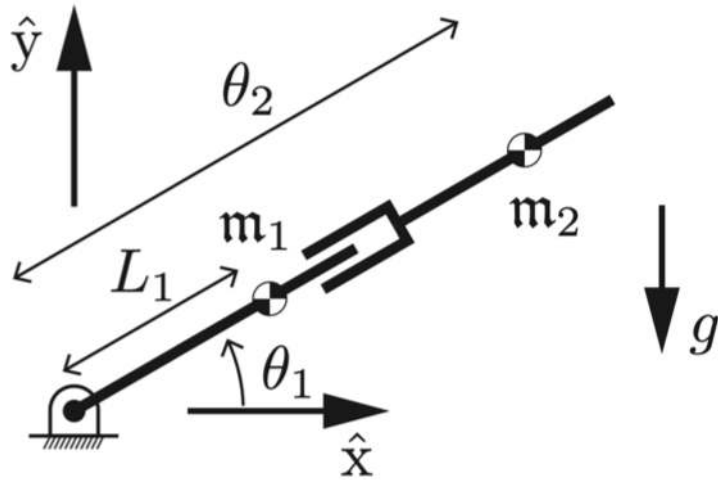
$$\mathcal{L} = \mathcal{K}_1 + \mathcal{K}_2 - \mathcal{P}_1 - \mathcal{P}_2 = \frac{1}{2} \mathfrak{m}_2 \theta_2^2 \dot{\theta}_1^2 + \dots$$

$$= \mathcal{L}_1 + \dots$$

product rule: $(fg)' = f'g + fg'$

chain rule: $\frac{df(g(t))}{dt} = \left(\frac{\partial f}{\partial g} \right) \frac{dg}{dt}$

RP in gravity



$$\mathcal{L}_1 = \frac{1}{2} m_2 \theta_2^2 \dot{\theta}_1^2$$

Contribution to τ_2 :

$$\begin{aligned} \tau_2 &= \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\theta}_2} - \frac{\partial \mathcal{L}}{\partial \theta_2} \\ &= 0 - m_2 \theta_2 \dot{\theta}_1^2 \end{aligned}$$

centripetal term

Contribution to τ_1 :

$$\tau_1 = \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\theta}_1} - \frac{\partial \mathcal{L}}{\partial \theta_1} \rightarrow 0$$

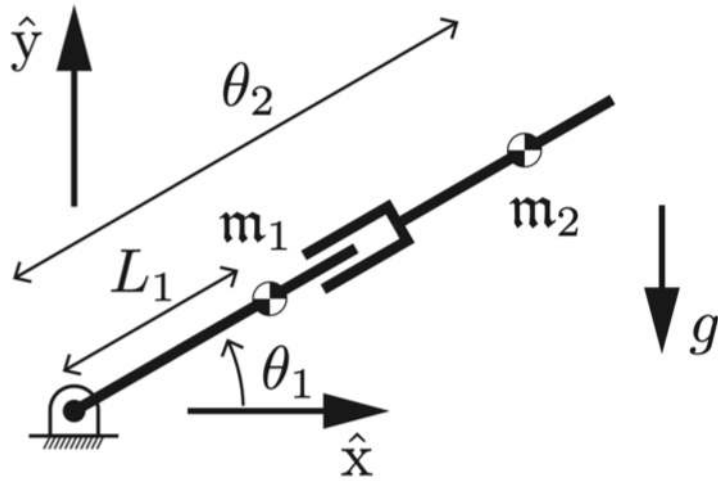
$$= \frac{d}{dt} (m_2 \theta_2^2 \dot{\theta}_1)$$

$$= 2 m_2 \theta_2 \dot{\theta}_1 \dot{\theta}_2 + m_2 \theta_2^2 \ddot{\theta}_1$$

Coriolis

mass matrix term

RP in gravity



$$\tau = M(\theta)\ddot{\theta} + c(\theta, \dot{\theta}) + g(\theta)$$

$$M(\theta) = \begin{bmatrix} \mathcal{I}_1 + \mathcal{I}_2 + m_1 L_1^2 + m_2 \theta_2^2 & 0 \\ 0 & m_2 \end{bmatrix}$$

$$c(\theta, \dot{\theta}) = \begin{bmatrix} 2m_2 \theta_2 \dot{\theta}_1 \dot{\theta}_2 \\ -m_2 \theta_2 \dot{\theta}_1^2 \end{bmatrix}$$

$$g(\theta) = \begin{bmatrix} (m_1 L_1 + m_2 \theta_2) g \cos \theta_1 \\ m_2 g \sin \theta_1 \end{bmatrix}$$

torque to keep joint 1 stationary in gravity

force to keep joint 2 stationary in gravity

Explain the velocity-product terms.

Assume $\ddot{\theta} = 0$, $g = 0$

$\dot{\theta}_2 = 0$: $\tau_2 = -m_2 \theta_2 \dot{\theta}_1^2$ is a force needed to prevent m_2 flying out tangent to the circle of motion

$\tau_1 = 2m_2 \dot{\theta}_1 \dot{\theta}_2$: Assume $\dot{\theta}_1 > 0$.
If $\dot{\theta}_2 < 0$: inertia of arm about joint 1 is decreasing, so $\tau_1 < 0$ to prevent $\dot{\theta}_1$ from increasing like a skater does (due to conservation of angular momentum) when she pulls her arms in